Balanzify Employee Tax Info UI, W-4 Generation & Payroll Compliance Logic

Federal + State + Local Tax Profile Design for Accurate Payroll Calculation

Field	Details
Document version	v1.0 - Employee Tax Info UI and W-4 generation specification
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Prepared for	Balanzify HR & Payroll product, engineering, QA, and payroll operations teams
Primary objective	Define a compliance-ready Tax Info tab that captures accurate employee tax certificates, dynamically applies state/local rules, generates Form W-4 correctly, and gives the payroll engine reliable calculation inputs.
Included jurisdictions	Federal baseline, New York/New York City/Yonkers implementation logic, New York employer-side MCTMT implications, and a reusable state-framework with Minnesota example logic.
Important implementation limit	No software document can guarantee permanent legal compliance by itself. Production accuracy depends on current official tables, employer-specific notices, employee-submitted forms, address history, year-to-date wages, and periodic review by qualified payroll tax professionals.

Executive guidance
The Tax Info screen should not be a generic form with manually editable state/city values. It should be a rule-driven tax profile built from employee identity, home address, work location, tax certificates, employer tax settings, and effective-dated official tax tables. Derived fields such as work state, work city, NYC resident flag, MCTMT zone, and tax jurisdiction should be read-only and changed only through the source record that owns the fact.

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Purpose and design principles

This document defines how Balanzify should redesign and implement the Employee Profile > Tax Info tab so payroll calculations, employee tax forms, tax previews, and generated Form W-4 outputs stay consistent. The goal is to prevent common payroll mistakes such as using one flat federal rate, applying New York City withholding only because the employee works in NYC, failing to update state forms, and allowing manually edited derived tax jurisdictions.

1.1 Core principles

Principle	Required product behavior
Source-of-truth driven	Legal name, SSN, home address, work location, employer EIN, and first date of employment must come from the owning profile records. Tax Info may display them but should not silently override them.
Federal and state forms are separate	Federal Form W-4 does not replace state withholding certificates. New York uses IT-2104/IT-2104-E/IT-2104.1. Minnesota uses W-4MN/MWR in applicable cases.
Derived jurisdictions are read-only	Work state/city and resident state/city should be computed from addresses and address effective dates. If work location is NYC, the screen should automatically show Work Locality = New York City and MCTMT Zone 1; payroll should not allow editing that field directly on Tax Info.
Residency controls resident local tax	NYC resident tax applies when the employee is an NYC resident. Work location in NYC alone should not enable NYC resident withholding.
Effective dating is mandatory	Every tax certificate, address, work location, exemption, and lock-in letter must have effective start and end dates so historical payroll remains reproducible.
Auditability over manual overrides	Manual override should be rare, permission-controlled, reason-required, and visible in payroll preview and audit logs.
Calculation snapshots	Finalized payroll must store the exact tax profile, rate version, formula inputs, taxable base, and YTD balances used at approval time.

Compliance wording
Use terms such as compliance-ready, audit-ready, and accuracy-controlled in the product documentation. Avoid promising permanent 100% compliance because payroll laws, tax tables, rate notices, and employee circumstances change. The system can only be accurate when current inputs and tables are maintained.

Current Tax Info screen gap analysis

The provided screens show a useful base: Federal Withholdings and State Other Income Tax sections already capture W-4 version, filing/withholding status, Step 2 multiple jobs, Step 3 dependents, Step 4(a), Step 4(b), Step 4(c), state of residence, state marital status, allowances, additional withholding, exemption, and a live/work-state question. To make the screen production-grade, the following gaps should be closed.

Current area	Risk / gap	Required improvement
State where employee lives	It can look manually editable and may not be tied to the address record.	Derive resident state from Home Address. Allow correction only through Address effective-dated update or approved residency override.
Does employee work in state where they live?	A Yes/No field can become stale after job transfer or remote work change.	Compute from Home Address and Work Location. Show as read-only: Same State / Different State / Multiple Work States.
Work location and NYC	Work location in NYC may be confused with NYC resident withholding.	Auto-populate work locality from Work Location. Use it for wage sourcing and employer taxes. Apply NYC resident withholding only when the resident address/local tax status indicates NYC resident.
Federal exempt from withholding	Exempt is shown as a dropdown but may not validate annual renewal or block contradictory fields.	When FIT Exempt = Yes, require employee certification, signed W-4, exemption year, and automatic expiration on February 15 of the next year. Disable Step 2/3/4 monetary entries unless a valid form requires otherwise.
Is this employee exempt from federal tax	This can be misunderstood as exempt from all federal payroll taxes.	Separate Federal Income Tax Exempt, FICA Exempt, FUTA applicability, and NRA/treaty handling. Exempt from FIT does not automatically stop Social Security or Medicare.
State allowances	Modern federal W-4 has no allowances, but some states still use allowances.	Use state-specific certificate logic. For New York, if 2020+ federal W-4 exists and no IT-2104 is filed, use zero allowances as required by NY guidance.
Generated W-4	The screen currently lacks signature, signed date, employer section, first employment date, and substitute-form recordkeeping controls.	Add W-4 Generation panel with employee signature/e-sign, date signed, employer name/address, EIN, first date of employment, generated PDF snapshot, and hardcopy export.
Audit trail	Tax changes affect payroll materially.	Every form edit must create an EmployeeTaxCertificate version. Past payroll should keep the old form version.

Recommended Tax Info tab structure

3.1 Top-level layout

Panel	Purpose	Key behavior
Tax Profile Summary	Shows current federal, state, local, and form status at a glance.	Read-only status badges: W-4 Active, IT-2104 Missing, NYC Resident, MCTMT Zone 1, Lock-in Letter Active, Exempt Expiring.
Location & Jurisdiction	Explains how resident and work tax jurisdictions were derived.	Home address drives resident state/locality. Work location drives work state/locality, employer taxes, wage sourcing, MCTMT zone, and UI state.
Federal Withholding - Form W-4	Captures the employee's active W-4 details.	Supports 2020+ W-4, legacy W-4 for existing records, exempt W-4, nonresident alien special instructions, lock-in letter controls, and generated form snapshot.
State Withholding	Shows state-specific fields based on resident and work state.	For New York show IT-2104/IT-2104-E/IT-2104.1 logic. For Minnesota show W-4MN/MWR logic. Unsupported states should show a configurable state form adapter.
Local Tax	Handles local resident/work withholding such as NYC or Yonkers.	Read-only eligibility determined by address lookup and state rules. Fields appear only when jurisdiction applies or is potentially applicable.
Employer-side Tax Impact	Shows taxes affected by the employee's work location and wages.	Display-only for employee profile: state UI, RSF, MCTMT covered employee/zone, FUTA, employer FICA. Actual rates remain in Employer Tax Settings.
Tax Form Documents	Stores generated and signed forms.	Versioned PDFs, source type, signed by, signature timestamp, IP/device metadata for e-sign, uploaded source document, and history.

3.2 Tax Profile Summary badges

Badge	When shown	User action
Federal W-4 Active	Active W-4 certificate exists for payroll date.	No action unless employee wants update.
Federal W-4 Missing	No valid federal W-4 exists.	Apply IRS default withholding and prompt HR/employee to complete W-4.
Federal Exempt	Active W-4 claims exemption from federal income tax withholding.	Show expiration date and require renewal workflow.
Lock-in Letter Active	IRS lock-in letter exists with an effective date.	Disable lower-withholding W-4 changes until release; allow higher withholding.
State Form Missing	A state certificate is required but not on file.	Apply state default rule and alert HR.
NYC Resident	Resident address is in NYC or approved residency override says NYC resident.	Enable NYC resident withholding fields and calculations.
Works in NYC	Work Location city/county maps to NYC Zone 1.	Show work tax sourcing and MCTMT Zone 1, but do not auto-enable NYC resident withholding.
Different Work/Resident State	Resident state differs from work state.	Show reciprocity, nonresident allocation, and multi-state withholding section.
Exemption Expiring	FIT/state exemption ends within configured warning window.	Prompt renewal before payroll runs after expiration.

Master location logic: home address vs work location

Critical compliance rule
The system must maintain two separate concepts: Resident Location and Work Location. Resident Location comes from the employee's home address and determines resident state/local taxes such as NYC resident withholding. Work Location comes from the employee's assigned job/worksite and determines wage sourcing, employer unemployment, and employer-side taxes such as MCTMT. Do not merge these concepts.

Location source	Owned by	Editable in Tax Info?	Payroll effect
Home Address	Contact tab / employee address history	No. Show read-only. Change by editing address with effective date.	Resident state, resident city, resident county, local resident tax eligibility, Form W-4 address, state certificate address.
Mailing Address	Contact tab	No. Show only if forms should use mailing address by policy.	Usually not used for tax jurisdiction unless selected as tax mailing address.
Work Location	Job tab / company location master	No. Show read-only. Change by assigning a different work location effective date.	Work state, work city, work county, state wage sourcing, employer UI state, MCTMT zone, local work taxes where applicable.
Remote/Hybrid Work Allocation	Time/work allocation profile	Only through approved work allocation workflow.	May split wages across work states/localities if state law and company policy require allocation.
Residency Override	Payroll/HR tax admin with approval	Only through controlled override drawer.	Used for part-year resident, temporary relocation, military spouse, or documented tax treatment that address alone cannot determine.

4.1 NYC example rule

Condition	Auto-populated field	Can user change on Tax Info?	Tax result
Work Location is 328 E 66th St, New York, NY 10065	Work State = NY; Work City = New York City; Work County/Borough = Manhattan/New York County; MCTMT Zone = Zone 1	No. Change the employee's Work Location record instead.	New York work-state sourcing and employer MCTMT Zone 1 evaluation. Does not by itself create NYC resident withholding.
Home Address is in Brooklyn, Manhattan, Queens, Bronx, or Staten Island	Resident City = New York City; NYC Resident = Yes	No. Correct address or add approved residency override.	Enable NYC resident withholding and include NYC tax in payroll.
Home Address is outside NYC but Work Location is NYC	NYC Resident = No; Works in NYC = Yes	No.	Do not withhold NYC resident income tax. Evaluate New York State withholding and MCTMT/employer rules.
Employee moved into or out of NYC mid-year	Create address/residency version with effective date	No direct overwrite.	Payroll after the effective date uses the new NYC status. Prior payroll remains locked under the old profile.

```
resolve_tax_location(employee, payroll_date):
    home = get_effective_home_address(employee.id, payroll_date)
    work = get_effective_work_location(employee.id, payroll_date)

    resident_state = normalize_state(home.state)
    resident_city = address_lookup(home).city_or_locality
    resident_county = address_lookup(home).county

    work_state = normalize_state(work.state)
    work_city = address_lookup(work.address).city_or_locality
    work_county = address_lookup(work.address).county

    nyc_resident = resident_state == 'NY' and resident_city in NYC_BOROUGHS_OR_LOCALITIES
    works_in_nyc = work_state == 'NY' and work_county in ['New York','Bronx','Kings','Queens','Richmond']
    mctmt_zone = get_mctmt_zone(work_county) if work_state == 'NY' else None

    return TaxLocationSnapshot(resident_state, resident_city, work_state, work_city, nyc_resident, works_in_nyc,
mctmt_zone)
```

Federal W-4 UI fields, validations, and payroll effects

5.1 Federal W-4 field specification

UI field	W-4 source	Required?	Payroll calculation effect	Validation / show-hide rule
W-4 form version	System metadata	Yes	Selects 2020+ method or legacy bridge method for old forms.	Default to current 2026 W-4 for new employees. Legacy W-4 only allowed for existing valid records.
Form effective date	System metadata	Yes	Determines first payroll date where this certificate applies.	Cannot be after payroll approval date unless creating a future certificate.
Legal first/middle/last name	Employee profile -> W-4 Step 1(a)	Yes	Identification and generated W-4 output.	Read-only here; edit in General profile.
SSN	Employee profile -> W-4 Step 1(b)	Yes	Identification and W-2/payroll reporting.	Masked after save. Full view only for authorized payroll roles.
Tax address	Home address -> W-4 Step 1 address	Yes	Generated W-4 and tax jurisdiction check.	Read-only here; edit in Contact/Address with effective date.
Filing status	W-4 Step 1(c)	Yes unless Exempt-only form	Controls Pub. 15-T table/column and annualized calculation.	Allowed values: Single or Married Filing Separately; Married Filing Jointly or Qualifying Surviving Spouse; Head of Household.
Multiple jobs/spouse works	W-4 Step 2(c)	Optional	Increases withholding through Pub. 15-T Step 2 treatment.	Show for 2020+ W-4. Disable when FIT Exempt = Yes.
Dependent credits	W-4 Step 3	Optional	Reduces annualized withholding.	Currency amount >= 0. Disable when FIT Exempt = Yes.
Other income	W-4 Step 4(a)	Optional	Increases annualized taxable amount for withholding.	Currency amount >= 0. Disable when FIT Exempt = Yes.
Deductions	W-4 Step 4(b)	Optional	Reduces annualized taxable amount for withholding.	Currency amount >= 0. Disable when FIT Exempt = Yes.
Extra withholding	W-4 Step 4(c)	Optional	Adds an exact amount to federal income tax withholding each pay period.	Currency amount >= 0. May remain allowed for lock-in/higher withholding scenarios.

UI field	W-4 source	Required?	Payroll calculation effect	Validation / show-hide rule
Claim exemption from federal income tax withholding	W-4 Exempt statement	Optional but controlled	Sets federal income tax withholding to $0 while active.	Require certification for prior and current year tax liability conditions; expires Feb 15 next year.
Nonresident alien indicator	IRS Notice 1392 / Pub. 15-T handling	Conditional	Applies NRA additional amount and special rules where applicable.	Show only if worker tax residency indicates nonresident alien or visa/treaty flags.
Lock-in letter active	IRS notice	Conditional	Overrides employee W-4 if employee requests lower withholding than lock-in.	Only payroll admin can enter. Require notice date, effective date, permitted status/credits/deductions.
Employee signature/e-signature	W-4 signature	Yes for generated form	Makes W-4 certificate valid for recordkeeping.	Capture signature, signed date, IP/device metadata for e-sign, and generated PDF snapshot.

5.2 Federal exempt logic

Condition	System action	Payroll effect
Employee checks federal income tax exemption for the current year	Require employee certification text, signed W-4, tax year, signed date, and effective date. Show warning that exemption applies only to federal income tax withholding.	Federal income tax withholding = $0. Continue Social Security and Medicare unless a separate FICA exemption exists.
Exempt certificate reaches February 15 of following year without renewal	Automatically mark exempt status expired before payroll after the deadline.	Withhold as Single or Married Filing Separately with no Step 2/3/4 entries until a new valid W-4 is received.
Exempt W-4 submitted after February 15	Apply to future payroll only.	Do not refund federal income tax withheld while no valid exempt status was in place.
Employee has IRS lock-in letter	Disable any employee W-4 that decreases withholding below lock-in terms.	Use lock-in parameters until IRS release is recorded. Higher withholding requests may be accepted.

5.3 Federal calculation formulas

```
fit_taxable_wages = gross_wages - pre_tax_deductions_that_reduce_fit
fit = pub15t_calculation(
    payroll_date, pay_frequency, fit_taxable_wages,
    filing_status, step2_multiple_jobs, step3_credits,
    step4a_other_income, step4b_deductions, step4c_extra_withholding,
    nonresident_alien_flag, lock_in_letter_if_any
)

if federal_income_tax_exempt_active:
    fit = 0

ss_current_taxable = min(ss_subject_wages, max(0, 184500 - ss_taxable_ytd_prior))
ss_employee = round(ss_current_taxable * 0.062, 2)
ss_employer = round(ss_current_taxable * 0.062, 2)

medicare_employee = round(medicare_subject_wages * 0.0145, 2)
medicare_employer = round(medicare_subject_wages * 0.0145, 2)

addl_medicare_base = max(0, medicare_ytd_prior + medicare_subject_wages - 200000) - max(0, medicare_ytd_prior -
200000)
additional_medicare_employee = round(addl_medicare_base * 0.009, 2)
```

Form W-4 generation mapping

Generation rule
Generate the W-4 from the locked EmployeeTaxCertificate snapshot, not from mutable live fields. This ensures the PDF exactly matches what payroll used at the time of payroll approval.

Generated Form W-4 field	Balanzify source field	Data owner	Validation
Step 1(a) First name and middle initial	employee.legal_first_name + middle_initial	General tab	Required. Use legal name, not display name.
Step 1(a) Last name	employee.legal_last_name	General tab	Required.

Generated Form W-4 field	Balanzify source field	Data owner	Validation
Step 1(b) Social Security number	employee.ssn	General tab / secure identity store	Required for payroll. Mask in UI but render full on official form if user has permission.
Address	effective_home_address.address_line_1 + line_2	Contact tab	Required.
City or town, state, ZIP code	effective_home_address.city/state/zip	Contact tab	Required. Must pass address normalization.
Step 1(c) Filing status checkbox	federal_w4.filing_status	Tax Info > Federal	One option only.
Step 2(c) Multiple jobs checkbox	federal_w4.multiple_jobs_or_spouse_works	Tax Info > Federal	Boolean.
Step 3 Claim dependents amount	federal_w4.step3_dependents_amount	Tax Info > Federal	Currency >= 0.
Step 4(a) Other income	federal_w4.step4a_other_income	Tax Info > Federal	Currency >= 0.
Step 4(b) Deductions	federal_w4.step4b_deductions	Tax Info > Federal	Currency >= 0.
Step 4(c) Extra withholding	federal_w4.step4c_extra_withholding	Tax Info > Federal	Currency >= 0 per paycheck.
Exempt statement	federal_w4.exempt_year + employee certification	Tax Info > Federal	Render exact exempt statement when active; block contradictory Step 2/3/4 entries.
Employee signature	tax_certificate.signature_hash / e-sign record	Tax Form Documents	Required for a valid generated form.
Date	tax_certificate.signed_at	Tax Form Documents	Required.
Employer name and address	company.legal_name + company.tax_address	Company tax profile	Required in employer section when generating complete employer copy.
First date of employment	employee.hire_date / first_date_of_employment	Job tab	Required for employer section.
Employer identification number	company.ein	Company tax profile	Required for employer section.

6.1 W-4 generation workflow

1. Load the effective EmployeeTaxCertificate for the selected payroll date.
2. Load employee legal identity, SSN, and effective home address from source-of-truth records.
3. Validate certificate state: active, signed, not expired, not superseded, and not blocked by lock-in letter.
4. Render Form W-4 with exact employee entries and employer section.
5. Hash the generated PDF and store the hash, generated time, user, certificate version, and source data version IDs.
6. Expose Download W-4, Replace W-4, and View History actions according to permissions.

State tax UI framework and dynamic show/hide rules

State UI layer	Required fields	Dynamic behavior
State selector	Resident state (derived), work state (derived), same/different state flag, state form type	Derived from address/work location. Direct user edits are disabled.
State certificate	Certificate type, tax year, effective date, signed date, filing/marital status, allowances or credits, additional withholding, exemption status	Fields are rendered from StateTaxAdapter for that state. New York shows IT-2104 fields. Minnesota shows W-4MN fields.
Reciprocity / nonresident	Reciprocity eligibility, nonresident allocation form, percentage of wages in work state, certificate upload	Only show when resident state differs from work state or state rules require allocation.
Local tax	Resident local jurisdiction, work local jurisdiction, local certificate if required	Only show applicable local tax forms. Example: NYC resident withholding; Yonkers resident/nonresident.
State exemptions	Exemption reason, form type, expiration, supporting document	Exemption should be certificate-based. Federal exemption does not imply state exemption.
State defaults	Default method when state form is missing	Each state adapter must define the legal default. Example: New York may use zero allowances for 2020+ W-4 with no IT-2104. Minnesota defaults to single/zero if no W-4MN.

7.1 Show/hide rule matrix

Trigger	Show	Hide / disable	Payroll effect
W-4 version = 2020 or later	Step 1(c), Step 2(c), Step 3, Step 4(a), Step 4(b), Step 4(c)	Federal allowances field	Use Pub. 15-T 2020+ W-4 method.
W-4 version = 2019 or earlier	Legacy filing status and allowances fields	2020+ Step 3/4 fields unless using computational bridge	Use valid legacy W-4 method or computational bridge according to employer configuration.
Federal income tax exempt = Yes	Exempt certification, tax year, expiration warning	Step 2/3/4 fields unless required for display history	FIT = 0 while active. FICA remains calculated separately.

Trigger	Show	Hide / disable	Payroll effect
NRA = Yes	NRA supplemental instructions, treaty/8233 fields, visa/residency review status	Standard-only W-4 validation	Apply IRS NRA withholding adjustments as required.
Resident state = NY or work state = NY	New York State withholding section and IT-2104 / IT-2104-E as applicable	Other state-specific form sections	Use NYS-50-T-NYS tables/methods and IT-2104/default rules.
Resident city = NYC	NYC resident withholding section	Manual NYC checkbox	Use NYS-50-T-NYC for NYC resident withholding.
Work location = NYC, resident city not NYC	Works in NYC badge, MCTMT Zone 1 display	NYC resident withholding section	No NYC resident withholding. Employer-side MCTMT evaluation may apply.
Resident or work city = Yonkers	Yonkers resident/nonresident withholding section	NYC fields unless also NYC resident is documented	Use NYS-50-T-Y where applicable.
Resident state differs from work state	Multi-state tax panel; reciprocity/nonresident forms; allocation	Same-state simple view	Calculate according to work/resident state rules and forms.
State = MN	W-4MN fields; MWR reciprocity if resident is MI/ND and work state MN	NY IT-2104 fields	Use Minnesota withholding tables/formula, W-4MN allowances, and reciprocity where documented.

New York + NYC + Yonkers withholding logic

8.1 New York employee withholding fields

UI field	NY form/source	When shown	Payroll effect
NY certificate type	IT-2104 / IT-2104-E / IT-2104.1	Resident state = NY, work state = NY, or NY withholding obligation detected	Determines which NY fields and defaults apply.
NY filing/marital status	IT-2104	NY withholding section	Used by NY withholding tables/methods where applicable.
NY total allowances	IT-2104 worksheet/line allowances	IT-2104 active	More allowances generally reduce NY withholding; zero allowances is conservative default when required.
Additional NYS withholding	IT-2104 additional amount	IT-2104 active	Adds extra NYS withholding per paycheck.
NY exemption	IT-2104-E or other NY exemption certificate	Employee claims exemption	Suppresses NY withholding only when valid certificate is active.
NYC resident flag	Address lookup / residency profile	Resident address in NYC or approved override	Enables NYC resident withholding.
Additional NYC withholding	IT-2104 city amount if supported	NYC resident	Adds extra NYC withholding per paycheck.
Yonkers resident flag	Address lookup / residency profile	Resident address in Yonkers or approved override	Enables Yonkers resident withholding.
Yonkers nonresident allocation	IT-2104.1 / Yonkers allocation	Work location or wage sourcing to Yonkers	Calculates Yonkers nonresident earnings tax where applicable.
Part-year city resident dates	Residency history / IT-360.1 support data	Moved into/out of NYC/Yonkers during year	Allows payroll to apply correct local withholding from effective dates.

8.2 New York calculation logic

```
ny_tax_profile = resolve_ny_profile(employee, payroll_date)
ny_taxable_wages = gross_wages - deductions_that_reduce_ny_withholding

if ny_tax_profile.ny_exempt_active:
    nys_withholding = 0
else:
    allowances = ny_tax_profile.it2104_allowances
    if allowances is None and federal_w4.version >= 2020:
        allowances = 0  # NY default if no IT-2104 is filed for 2020+ federal W-4
    nys_withholding = nys50_t_nys_calculation(payroll_date, pay_frequency, ny_taxable_wages, ny_tax_profile.status,
allowances)
    nys_withholding += ny_tax_profile.additional_nys_withholding

if tax_location.nyc_resident:
    nyc_withholding = nys50_t_nyc_calculation(payroll_date, pay_frequency, ny_taxable_wages,
ny_tax_profile.nyc_fields)
    nyc_withholding += ny_tax_profile.additional_nyc_withholding
else:
    nyc_withholding = 0

if ny_tax_profile.yonkers_applicable:
    yonkers_withholding = nys50_t_y_calculation(payroll_date, pay_frequency, yonkers_subject_wages,
ny_tax_profile.yonkers_fields)
```

```
else:
    yonkers_withholding = 0
```

New York warning
NYC resident withholding is based on NYC residency. If the employee works at a New York City office but lives outside NYC, do not withhold NYC resident tax. The employee may still have New York State withholding and the employer may still evaluate MCTMT based on work location and employer covered payroll.

Employer-side location-driven taxes: NY UI, RSF, MCTMT, PFL, DBL

Tax / deduction	Employee or employer	Where configured	Tax Info screen behavior
NY UI / SUI	Employer	Company Employer Tax Profile	Display state UI applicability based on work state. Rate is not employee-editable. Use employer annual rate notice and wage base ledger.
NY Re-employment Service Fund	Employer	Company Employer Tax Profile / system rate table	Display separately from UI. Do not blend into one rate for audit.
MCTMT	Employer	Company MCTMT settings + work location zone	Show covered employee candidate and zone from work location. Final liability is employer-quarter level, not per employee withholding.
NY Paid Family Leave	Usually employee-funded unless employer pays	Payroll deduction/tax settings	Show whether deduction applies and annual cap status. It is not part of Form W-4.
NY Disability Benefits deduction	Employee if employer collects	Payroll deduction/tax settings	Show DBL collection setting and weekly cap. It is not part of Form W-4.
FUTA	Employer	Company Federal Tax Profile	Display-only. Based on employer and wage base ledger.
Employer FICA	Employer	Federal tax engine	Display-only. Matches employee Social Security/Medicare taxable base except Additional Medicare has no employer match.

9.1 MCTMT UI behavior

Work location county	MCTMT zone	Tax Info display	Calculation note
New York/Manhattan, Bronx, Kings/Brooklyn, Queens, Richmond/Staten Island	Zone 1	Works in MCTD - Zone 1	Use employer-quarter covered payroll threshold and Zone 1 rates.
Rockland, Nassau, Suffolk, Orange, Putnam, Dutchess, Westchester	Zone 2	Works in MCTD - Zone 2	Use employer-quarter covered payroll threshold and Zone 2 rates.
Outside MCTD	None	Not MCTMT-covered from this work location	No MCTMT payroll expense for this work location unless allocation/residency test creates coverage.

Minnesota example implementation

Minnesota is included as an example state adapter because your uploaded references include Federal + Minnesota 2026 payroll tax documentation. This structure can be reused for other states by replacing the state adapter fields, forms, defaults, and withholding formulas.

Minnesota UI field	Source/form	When shown	Payroll effect
MN certificate type	W-4MN	Resident state = MN or work state = MN	Determines Minnesota withholding allowances or exemption.
MN filing status	W-4MN	W-4MN active	Used by Minnesota withholding tables/formula.
MN allowances	W-4MN Section 1	W-4MN active	Allowance count used in Minnesota regular wage withholding.

Minnesota UI field	Source/form	When shown	Payroll effect
Additional MN withholding	W-4MN additional amount	W-4MN active	Adds extra Minnesota withholding per pay period.
MN exemption	W-4MN Section 2	Employee claims exemption	Suppresses Minnesota withholding only while valid.
MWR reciprocity	Form MWR	Employee lives in Michigan or North Dakota and works in Minnesota	Suppress Minnesota withholding when annual form is valid and reciprocity applies.
MN Paid Leave	State program settings	Work/resident rules require Minnesota Paid Leave premium	Apply employee/employer premium shares according to employer settings and wage cap.
MN UI	Employer UI rate notice	Work state = MN and covered employment	Employer-only; never deduct from employee pay.

```
if state_adapter == 'MN':
    if no_valid_w4mn:
        mn_status = 'Single'
        mn_allowances = 0
    if reciprocity_form_mwr_active and resident_state in ['MI', 'ND'] and work_state == 'MN':
        mn_withholding = 0
    else:
        mn_withholding = mn_withholding_formula(pay_frequency, mn_taxable_wages, mn_status, mn_allowances)
        mn_withholding += additional_mn_withholding

    mn_ui = employer_only(company_mn_ui_rate, min(mn_ui_wages, remaining_mn_ui_wage_base))
    mn_paid_leave_employee = calculate_mn_paid_leave_employee_share(...)
    mn_paid_leave_employer = calculate_mn_paid_leave_employer_share(...)
```

Payroll calculation order from Tax Info values

Step	Action	Tax Info dependency	Validation
1	Load employee and payroll date	Employee status, effective Tax Info certificate, home address, work location	Employee must be active or payable in the period.
2	Resolve tax location	Home address + work location + residency override	Derived state/local fields must match source records.
3	Calculate gross earnings	Pay type, salary/hourly, overtime, bonus, commission	Separate regular and supplemental wages.
4	Apply pre-tax deductions	Deduction taxability matrix	Each deduction must specify which tax bases it reduces.
5	Federal income tax	Active W-4 or default rule, lock-in letter	Use Pub. 15-T. Do not apply annual bracket directly.
6	FICA	FICA subject wages and YTD	Stop Social Security at wage base; start Additional Medicare after threshold.
7	State withholding	State certificate, resident/work state, state adapter	Use state-specific default if certificate missing.
8	Local withholding	Resident/work local flags	NYC resident tax only for NYC resident; Yonkers if applicable.
9	Employee payroll deductions	PFL/DBL/benefits/garnishment settings	Apply caps and legal limits.
10	Employer taxes	Work location, employer tax profile, YTD wage-base ledger	Employer taxes do not reduce net pay.
11	Create preview and audit snapshot	All calculation inputs	Show taxable base, rate, formula, YTD before/after, and source.
12	Finalize payroll	Locked certificate and rate versions	No silent mutation after approval. Use void/reversal/adjustment.

Data model, audit trail, APIs, and security

12.1 Recommended entities

Entity	Critical fields	Purpose
EmployeeTaxProfile	employee_id, active_certificate_ids, resident_state, work_state, resident_city, work_city, local_tax_flags, effective_start, effective_end	Current tax profile summary used by payroll preview.

Entity	Critical fields	Purpose
EmployeeTaxCertificate	id, employee_id, form_type, form_year, jurisdiction, source_type, signed_at, effective_start, effective_end, superseded_by, status, certificate_json	Versioned W-4, IT-2104, W-4MN, exemption, nonresident, and local certificates.
AddressHistory	employee_id, address_type, line1, city, county, state, zip, effective_start, effective_end, verified_by, geocode_result	Determines resident jurisdiction and W-4 address.
WorkLocationAssignment	employee_id, location_id, allocation_percent, effective_start, effective_end	Determines work state/city, employer taxes, MCTMT zone, and multi-state allocation.
CompanyTaxProfile	company_id, EIN, legal_name, tax_address, deposit_schedule, federal_settings, state_accounts	Employer section on W-4 and company-level tax calculation settings.
StateTaxAdapter	state, supported_forms, default_rules, withholding_method, reciprocity_rules, local_tax_rules	Configurable state logic to avoid hard-coding all states in UI.
TaxRateVersion	jurisdiction, tax_type, tax_year, effective_start, effective_end, rate, wage_base, cap, table_version, official_source_url	Versioned official tax constants and tables.
PayrollTaxSnapshot	payroll_run_id, employee_id, tax_profile_json, rates_json, taxable_bases_json, formulas_json, created_at, locked_at	Audit proof of calculation inputs.
TaxFormDocument	certificate_id, file_id, generated_pdf_hash, signed_pdf_hash, created_by, created_at	Stores generated W-4/state forms and uploaded documents.
ManualTaxOverride	field, old_value, new_value, reason, attachment, requested_by, approved_by, timestamp	Controls exceptional edits.

12.2 API payload example

```
POST /employees/{employeeId}/tax-certificates/federal-w4
{
  "formYear": 2026,
  "formVersion": "2020_PLUS",
  "effectiveStart": "2026-01-01",
  "filingStatus": "SINGLE_OR_MARRIED_FILING_SEPARATELY",
  "multipleJobsOrSpouseWorks": false,
  "step3DependentsAmount": 0,
  "step4aOtherIncome": 0,
  "step4bDeductions": 0,
  "step4cExtraWithholding": 0,
  "federalIncomeTaxExempt": false,
  "nonresidentAlien": false,
  "signedAt": "2026-01-03T10:20:00Z",
  "signatureMethod": "EMPLOYEE_PORTAL_E_SIGN"
}
```

12.3 Security controls

- Mask SSN by default; reveal only to authorized payroll/HR roles with reason logging.
- Encrypt SSN, bank data, and signed tax form files at rest and in transit.
- Separate employee self-service permissions from HR/payroll admin permissions.
- Require approval for overrides that reduce withholding or change residency/local tax status.
- Keep tax forms for the required retention period and be able to export a hardcopy of electronic Form W-4.
- Log every change with old value, new value, user, timestamp, IP/device metadata, effective date, and source form/document.

UI interaction rules and validation matrix

Validation	Blocking?	Message / behavior
No valid W-4 on file	Warning, not always blocking	Use IRS default withholding and prompt completion.
W-4 exempt selected without certification	Blocking	Employee must certify prior-year and current-year tax liability conditions and sign the form.
Federal exempt plus Step 2/3/4 amounts	Blocking or controlled warning	Exempt certificate should not carry contradictory withholding entries.
FIT exempt but FICA exempt not documented	Informational	Federal income tax withholding stops; Social Security and Medicare continue.
State form missing where required	Warning/blocking based on payroll policy	Apply state default rule and alert HR/employee.
Home address edited after payroll already finalized	Blocking for historical mutation	Create new address version; do not mutate prior payroll location snapshot.
Work location edited from Tax Info	Blocking	Change Work Location in Job tab with effective date.
Manual NYC resident checkbox change	Blocking	NYC resident status must come from address lookup or approved residency override.

Validation	Blocking?	Message / behavior
Different resident/work state	Warning	Show reciprocity/nonresident allocation section.
Additional withholding negative	Blocking	Must be zero or positive currency.
Lock-in letter active and employee requests lower withholding	Blocking	IRS lock-in letter controls until release.
Exemption expiring within 30 days	Warning	Prompt employee to submit renewal before February 15.
Unsupported state adapter	Blocking for live payroll	Payroll cannot be finalized until tax adapter/rule is configured or state is marked not subject.

QA test cases

Test case	Input	Expected result
Federal W-4 missing	New employee has no W-4	Payroll defaults to Single/MFS with no Step 2/3/4 entries and shows missing W-4 warning.
2020+ W-4 normal	Single, Step 2 No, Step 3 = 0, Step 4 values = 0	FIT uses Pub. 15-T method with selected pay frequency.
Extra withholding	Step 4(c) = $25	Calculated FIT + $25 on each paycheck.
Federal exempt active	Exempt year 2026 signed W-4	FIT = $0; Social Security and Medicare still calculate.
Federal exempt expired	2025 exempt W-4 used for payroll after Feb 15, 2026 with no renewal	Exemption inactive; use default withholding unless new W-4 exists.
Lock-in letter	Lock-in requires Single/MFS, lower credits than employee request	System uses lock-in settings and blocks lower withholding changes.
Work NYC, live outside NYC	Work location New York, NY 10065; home address outside NYC	Works in NYC/MCTMT Zone 1 appears. NYC resident withholding remains disabled.
Live NYC	Home address in Manhattan/Brooklyn/Queens/Bronx/Staten Island	NYC resident withholding enabled and generated payroll includes NYC tax.
NY 2020+ W-4 but no IT-2104	Federal W-4 exists; no NY certificate	NY allowances default to zero and warning is shown.
NY exemption	IT-2104-E active	NY withholding suppressed only for active exemption and jurisdiction covered by certificate.
Yonkers resident	Home address in Yonkers	Yonkers resident section appears and withholding uses Yonkers method.
Minnesota no W-4MN	MN employee missing W-4MN	Minnesota withholding defaults to Single/zero allowances and warning is shown.
Minnesota reciprocity	MI resident works in MN with active MWR	MN withholding suppressed according to reciprocity rule.
Address effective date	Employee moves into NYC mid-month	Payroll before effective date uses old local status; payroll after effective date uses NYC status.
Generated W-4	Signed W-4 certificate with employee profile data	PDF contains legal name, SSN, address, filing status, steps, signature/date, employer name/address, first date of employment, EIN.

Product implementation checklist

15.1 Must-have before production

- Separate federal W-4 certificate from state withholding certificates.
- Make home address, work location, resident city, work city, and MCTMT zone effective-dated.
- Make resident/work jurisdictions read-only in Tax Info and derived from source records.
- Implement 2020+ W-4 fields and generated W-4 mapping with employee signature and employer section.
- Implement federal exempt renewal/expiration by February 15.
- Implement IRS lock-in letter support.
- Implement New York IT-2104, IT-2104-E, IT-2104.1, NYC resident, and Yonkers rules.
- Prevent NYC resident withholding from being triggered only by NYC work location.
- Implement state adapter framework for other states and at least Minnesota W-4MN as a reusable example.
- Store TaxRateVersion and PayrollTaxSnapshot for every approved payroll.
- Show payroll preview with formula, tax base, rate, YTD before/after, and source version.
- Require manual override approvals and full audit logs.

15.2 Nice-to-have improvements

- Address lookup integration to confirm city/county/local tax jurisdiction.
- Employee self-service guided W-4 and state certificate wizard.

- Form expiry dashboard for W-4 exempt, state exemption, and reciprocity certificates.
- Tax profile comparison view showing old vs new withholding effect before saving.
- Payroll simulation mode to preview net pay impact before approval.
- Official-source update workflow to import tables and archive old versions.

Official source register

Source	URL	Used for
IRS Topic No. 753 - Form W-4	https://www.irs.gov/taxtopics/tc753	W-4 field purpose, default withholding, revised W-4 timing, electronic systems, exemption validity, recordkeeping, invalid W-4, and lock-in letter guidance.
IRS Form W-4 2026	https://www.irs.gov/pub/irs-pdf/fw4.pdf	Generated W-4 field mapping and exempt statement.
IRS Publication 15-T 2026	https://www.irs.gov/publications/p15t	Federal income tax withholding methods, wage bracket, percentage method, and alternative methods.
IRS Publication 15 2026	https://www.irs.gov/publications/p15	Employer federal withholding, deposit, reporting, supplemental wage, FICA, and FUTA rules.
New York IT-2104 2026	https://www.tax.ny.gov/pdf/current_forms/it/it2104_fill_in.pdf	New York employee withholding certificate and annual review instructions.
New York IT-2104 Instructions 2026	https://www.tax.ny.gov/forms/current-forms/it/it2104i.htm	New York allowances and employee certificate guidance.
New York withholding amount guidance	https://www.tax.ny.gov/bus/wt/amount_deduct.htm	Approved withholding tables, address-based withholding, IT-2104 and IT-2104.1 usage, and zero allowance default for 2020+ W-4 without IT-2104.
New York NYS-50-T-NYS 2026	https://www.tax.ny.gov/pdf/publications/withholding/nys50_t_nys.pdf	New York State withholding tables, methods, and supplemental rate.
New York NYS-50-T-NYC 2026	https://www.tax.ny.gov/pdf/publications/withholding/nys50_t_nyc.pdf	New York City resident withholding tables and methods.
New York NYS-50-T-Y 2026	https://www.tax.ny.gov/pdf/publications/withholding/nys50_t_y.pdf	Yonkers resident and nonresident withholding tables and methods.
New York MCTMT employer guidance	https://www.tax.ny.gov/bus/mctmt/emp.htm	MCTMT threshold, zones, rates, and due dates.
NY DFS 2026 Paid Family Leave rate decision	https://www.dfs.ny.gov/apps-and-licensing/health-insurers/pfl-rate-decision-2026-page	2026 PFL contribution rate and annual cap.
NY Workers' Compensation Board Disability Benefits	https://www.wcb.ny.gov/content/main/DisabilityBenefits/employee-disability-benefits.jsp	DBL employee contribution rate and weekly cap.
Minnesota Form W-4MN	https://www.revenue.state.mn.us/form-w-4mn	Minnesota state certificate requirement and default withholding rule.
Minnesota 2026 Withholding Instructions	https://www.revenue.state.mn.us/sites/default/files/2025-12/wh-inst-26.pdf	Minnesota withholding tables and forms.
Minnesota UI tax rates	https://www.uimn.org/employers/wages-taxes/tax-rates/index.jsp	Minnesota UI wage base and employer-specific rate notice handling.

Appendix A - Developer implementation notes

Module	Implementation note
Frontend	Use a state adapter registry to render jurisdiction-specific fields. Do not use one massive hard-coded Tax Info form.
Backend	Use effective-dated certificates and address/work-location records. Payroll calculation should request a TaxLocationSnapshot and EmployeeTaxSnapshot for the payroll date.
QA	Test by jurisdiction, form version, missing form defaults, exemptions, address changes, and payroll effective dates.
Accounting	Keep employee tax liabilities, employer tax liabilities, wage payable, and deduction payable accounts separate for reconciliation.
Compliance operations	Review official IRS/state sources before annual first payroll and whenever states issue mid-year table changes.

Final production note
This document is an implementation specification, not legal or tax advice. Before live payroll filing, validate the actual engine against official federal and state tables, employer rate notices, employee certificates, and a qualified U.S. payroll tax advisor.